

Delta Drills — Current Mastery & Selection Model (EWMA stack with prerequisite gating)

Snapshot of the live engine as of 2026-05-26. Combines the original per-subtopic Exponentially Weighted Moving Average (EWMA) update rule with the prerequisite-gating layer that has been added since the original math document was written. The model is implemented across `backend/app/adaptive.py`, `backend/app/prioritization.py`, `backend/app/practice/arena_rating_router.py`, and the frontend `practice/arena-unlock.js` + `practice/drills-catalog.js`.

0. Notation

Let

- $\mathcal{S}$ — the finite set of subtopics covered by the question bank (e.g. *Einops: Rearrange*, *CNN: Conv output shape*).
- $\mathcal{T}$ — the finite set of topics. Every subtopic s belongs to exactly one topic $T(s) \in \mathcal{T}$.
- $\mathcal{D}$ — the catalog of procedural drills (Colab notebooks). Each drill $d \in \mathcal{D}$ carries
- $\text{subs}(d) \subseteq \mathcal{S}$ — the subtopics it exercises (one for single-atom drills, multiple for composites),
- a flag $\text{comp}(d) \in \{0, 1\}$ — whether d is a composite drill,
- an index $i(d) \in \mathbb{N}$ giving its position in the catalog list.
- $\mathcal{E}$ — the set of ARENA exercises. Each $e \in \mathcal{E}$ carries problem–concept links $L_e = \{(s_i, \omega_i)\}_{i=1}^{m_e}$ where $\omega_i \geq 0$ are the per-link skill weights.

For each subtopic s and student, the per-subtopic adaptive state is the triple

$$\sigma_s = (n_s, b_s, p_s) \in \mathbb{N} \times [0, 100] \times [0, 1],$$

where n_s is the number of recorded attempts, b_s is the running baseline score, and p_s is the running correctness rate. A timestamp τ_s of the last update is also kept for time decay (see §1.6).

1. Per-subtopic adaptive engine

This is the layer that turns each individual attempt into an update of σ_s and a target difficulty for the next question on that subtopic.

1.1 Cold start ($n_s \leq 3$)

The first three attempts on each subtopic are scripted to sample three difficulty bands:

$$\text{target}(n_s) = \begin{cases} 25 & n_s = 0 \\ 50 & n_s = 1 \\ 75 & n_s = 2 \end{cases} \quad (\text{cold-start ladder})$$

The state σ_s still accumulates during cold start (using the same EWMA blends as below), but the target difficulty is fixed by the ladder rather than computed from σ_s .

1.2 Indicator and feedback recency factor

After every attempt, the student self-reports a grade $\text{grade}(n) \in [0, 100]$ and selects a feedback level. The grade-binarizing indicator is

$$\mathbf{1}(\text{grade}(n)) = \begin{cases} 1 & \text{grade}(n) > 85 \\ 0 & \text{grade}(n) \leq 85 \end{cases}$$

The feedback level is mapped to a per-attempt recency factor $\alpha(n)$:

$$\alpha(n) = \begin{cases} 0.30 & \text{"Not much"} \\ 0.60 & \text{"Somewhat"} \\ 0.85 & \text{"A lot"} \end{cases}$$

A higher $\alpha(n)$ means the EWMA absorbs the current attempt more aggressively.

1.3 Correctness rate update

The running correctness rate evolves as a fixed-recency EWMA on the binarized indicator:

$$p(n) = \alpha(n) \cdot \mathbf{1}(\text{grade}(n)) + (1 - \alpha(n)) p(n - 1), \quad p(0) = 0.5.$$

1.4 Score and baseline

Each attempt has a difficulty $\text{diff}(n) \in [0, 100]$ assigned by the recommender. The raw score is

$$\text{score}(n) = \frac{\text{grade}(n) \cdot \text{diff}(n)}{100}.$$

The baseline (a running estimate of "how high a difficulty the student is consistently scoring at") is then

$$b(n) = \alpha(n) \cdot \text{score}(n) + (1 - \alpha(n)) b(n - 1), \quad b(0) = 0.$$

1.5 Difficulty multiplier and target difficulty

Define the piecewise multiplier

$$\mu(p) = \begin{cases} 0.5 + 0.5 \left(\frac{p}{0.85} \right)^{1.8} & p \leq 0.85 \\ \min \left(2.5, 1 + \left(\frac{p - 0.85}{0.15} \right)^{2.5} \right) & p > 0.85 \end{cases}$$

After the third attempt ($n_s > 3$), the target difficulty for the next problem on s is

$$\text{target}(n) = \text{clip}_{[0,100]} \left(b(n) \cdot \mu(p(n)) \right).$$

1.6 Time decay between attempts

Let Δt be the number of days elapsed since the last update timestamp τ_s , and let $\tau_{1/2} = 14$ days be the half-life. Just before applying the EWMA update of §1.3–1.5 on the next attempt, both b and p are shrunk toward their priors $B_0 = 0$ and $P_0 = 0.5$:

$$\begin{aligned} b &\leftarrow B_0 + (b - B_0) \cdot 2^{-\Delta t / \tau_{1/2}}, \\ p &\leftarrow P_0 + (p - P_0) \cdot 2^{-\Delta t / \tau_{1/2}}. \end{aligned}$$

This makes prolonged inactivity regress the mastery estimate back toward "no information" rather than letting it stay at the last observed value indefinitely.

2. Subtopic selection (next-question routing)

This layer decides *which subtopic to pull the next question from* whenever the student requests a new question.

2.1 User-defined effective weights

Through the Statistics page, the student assigns a topic percentage $P_T \in [0, \infty)$ to each topic $T \in \mathcal{T}$ and, within each topic, a subtopic share $P_s \in [0, \infty)$ to each s with $T(s) = T$. The effective weight of subtopic s is

$$w_s = \frac{P_{T(s)}}{100} \cdot \frac{P_s}{100}.$$

Weights need not sum to 1; only their relative magnitudes affect selection. If no custom weights are set, the system falls back to uniform weights

$$w_s = \frac{1}{|\mathcal{S}|}.$$

2.2 Learning-rate estimate via EWMA on baseline deltas

Let the recorded baseline sequence for subtopic s be $b_{s,1}, b_{s,2}, \dots, b_{s,n_s}$. Define the step deltas

$$\delta_{s,t} = b_{s,t} - b_{s,t-1}, \quad t = 2, \dots, n_s.$$

The learning-rate estimate $\hat{r}_s$ is an EWMA over these deltas with smoothing constant

$$\alpha_{\text{lr}} = 1 - e^{-\lambda}, \quad \lambda = 0.3,$$

(note: α_{lr} here is the *learning-rate EWMA smoothing*, distinct from $\alpha(n)$ in §1.2). For $n_s < 2$ there are no deltas yet, so we use a cold-start prior:

$$\hat{r}_s = 0.5 \quad \text{when} \quad n_s < 2.$$

Otherwise recursively define

$$\begin{aligned} s_1 &= \delta_{s,2} \\ s_k &= \alpha_{\text{lr}} \delta_{s,k+1} + (1 - \alpha_{\text{lr}}) s_{k-1}, \quad k = 2, \dots, n_s - 1 \end{aligned}$$

and set

$$\hat{r}_s = s_{n_s-1}.$$

2.3 Gradient, masking, and selection

Let $\mathcal{S}_{\text{open}} \subseteq \mathcal{S}$ be the set of subtopics with at least one unserved question. The subtopic priority (gradient) is

$$g_s = w_s \cdot \hat{r}_s.$$

The next subtopic is chosen by

$$s^* = \arg \max_{s \in \mathcal{S}_{\text{open}}} g_s.$$

If $\mathcal{S}_{\text{open}} = \emptyset$ (i.e. every question has been served), the served-set is reset and the selection is recomputed. Topics are not selected directly: the induced topic is simply

$$T^* = T(s^*).$$

The actual question is then drawn from the bank for subtopic s^* at a difficulty close to $\text{target}(n_{s^*})$ from §1.5.

3. Prerequisite-gated procedural-drill recommender

This is the layer that decides, *after* a question/feedback cycle finishes, whether to fire a procedural Colab drill — and if so, which one.

3.1 Per-subtopic mastery readout

The frontend's gating logic reads a scalar mastery score per subtopic from the backend state. With p_s the running correctness rate of §1.3, the readout is

$$\rho_s = 100 \cdot p_s \in [0, 100].$$

(For sake of completeness: if p_s is missing but a positive baseline exists, the readout falls back to b_s . If neither is set, ρ_s is undefined and treated as "no information.")

3.2 Drill unlock predicate

Each drill $d \in \mathcal{D}$ unlocks once every one of its constituent subtopics has crossed the unlock floor $\theta_{\text{unlock}} = 50$:

$$U(d) \equiv (\forall s \in \text{subs}(d) : \rho_s \text{ is defined and } \rho_s \geq \theta_{\text{unlock}}).$$

For single-atom drills this is one inequality; for composites it is a conjunction over 2–3 atoms.

3.3 Composite promote predicate

To prevent composite drills from never firing (they sit at the back of the catalog list, behind hundreds of single-atom drills), composites are *promoted* ahead of further single-atom drills once their atoms are strongly mastered. Let $\theta_{\text{promote}} = 70 > \theta_{\text{unlock}}$. Define

$$P(d) \equiv \text{comp}(d) \wedge (\forall s \in \text{subs}(d) : \rho_s \text{ is defined and } \rho_s \geq \theta_{\text{promote}}).$$

3.4 Two-pass selection

Let $\Sigma \subseteq \mathcal{D}$ be the set of drills already shown to the student (tracked in `localStorage`). The next drill is chosen in two passes:

Pass 1 (composite promotion).

$$d_1^* = \arg \min_d \{ i(d) : d \notin \Sigma \wedge P(d) \}.$$

Pass 2 (in-order unlocked fallback). If Pass 1 returns no drill,

$$d_2^* = \arg \min_d \{ i(d) : d \notin \Sigma \wedge U(d) \}.$$

The fired drill is $d^* = d_1^*$ if defined, else d_2^* ; if both are empty the recommender returns "no drill available."

3.5 ARENA exercise readiness

ARENA exercises live on a separate surface (the Courses tab). Each exercise $e \in \mathcal{E}$ has skill-weight links $L_e = \{(s_i, \omega_i)\}$, and its readiness score is the weight-normalized average of constituent atom masteries:

$$R(e) = \begin{cases} \frac{\sum_{(s_i, \omega_i) \in L_e} \omega_i \cdot \min(\rho_{s_i}, 100)}{\sum_{(s_i, \omega_i) \in L_e} \omega_i} & \text{if } L_e \neq \emptyset \\ R_0(e) & \text{otherwise (static fallback)} \end{cases}$$

$R(e)$ governs the readiness bar shown on ARENA cards and is consumed by **targeted-practice** for ordering search results.

4. Composite drill rating (multi-subtopic credit)

When the student rates a procedural drill, the system fires *one update per constituent subtopic*. This is the "even-skill averaging" credit-assignment scheme.

4.1 Self-rating mapping

Drill self-rating r is one of four buttons:

Button	Correct $c(r)$	Feedback $f(r)$
✓ Solved in target time, no help	1	"A lot"
✓ Solved in target time, with a hint	1	"Somewhat"
✓ Solved correctly, over target time	1	"Not much"
✗ Looked up the solution	0	"A lot"

The feedback level $f(r)$ maps to $\alpha(n) \in \{0.3, 0.6, 0.85\}$ exactly as in §1.2.

4.2 Parallel apply-feedback

For composite drill d with $\text{subs}(d) = \{s_1, \dots, s_k\}$, the rating event triggers k synthetic attempts in parallel. For each s_i :

$$\text{diff}_{s_i} = \lfloor \text{target}(n_{s_i}) \rfloor \quad (\text{fallback to 50 if undefined}),$$

$$\text{grade}_{s_i} = 100 \cdot c(r),$$

$$\text{score}_{s_i} = \frac{\text{grade}_{s_i} \cdot \text{diff}_{s_i}}{100}.$$

Then the §1.3–1.5 updates fire independently on each s_i with the same $\alpha(n) = \alpha(f(r))$. After the parallel updates,

$$n_{s_i} \leftarrow n_{s_i} + 1 \quad \forall i,$$

so each constituent atom advances one cold-start step. Each subtopic accumulates its own evidence; there is no shared per-composite latent state.

This is mathematically equivalent to "equal-credit" multi-tag PFA (cf. Maier 2021): each tagged subtopic receives the same correctness signal weighted by its own current target difficulty.

5. Constants summary

Symbol	Value	Source / meaning
$\alpha(n)$ ("Not much"/"Somewhat"/"A lot")	0.30 / 0.60 / 0.85	Per-attempt EWMA recency factor (§1.2)
α_{lr}	$1 - e^{-0.3} \approx 0.259$	Learning-rate EWMA smoothing (§2.2)
$\tau_{1/2}$	14 days	Mastery half-life (§1.6)
B_0, P_0	0, 0.5	Priors for baseline and correctness rate
Cold-start ladder	$\{25, 50, 75\}$	Target difficulties for $n_s \in \{0, 1, 2\}$
Grade threshold for "correct"	85	Indicator $\mathbf{1}(\text{grade} > 85)$
θ_{unlock}	50	Drill unlock floor (§3.2)
θ_{promote}	70	Composite promotion floor (§3.3)
$\mu(p)$ cap	2.5	Maximum difficulty multiplier (§1.5)
$\mu(p)$ knee	$p = 0.85$	Piecewise breakpoint (§1.5)

6. Composition diagram

The four layers compose top-to-bottom on every interaction:

- Subtopic selection (§2)** chooses s^* using $g_s = w_s \hat{r}_s$.
- Per-subtopic engine (§1)** serves a question at difficulty $\approx \text{target}(n_{s^*})$, then on feedback updates $\sigma_{s^*} = (n, b, p)$ with EWMA + time decay.
- Prereq-gated drill recommender (§3)** checks every drill against ρ_s readouts; fires a composite if any is promotable, otherwise the next in-order unlocked single-atom drill.

4. **Composite drill rating (§4)** propagates one rating into k parallel per-subtopic updates, each running the same §1 machinery.

There is no global "student ability" scalar; mastery lives per subtopic, and harder content unlocks through prerequisite conjunctions in §3, not through a global difficulty axis.

Engine code: *This-Directory-Only/backend/app/{adaptive.py, prioritization.py, practice/arena_rating_router.py}* (Python) and *Local_Deployed_Shared/{practice/arena-unlock.js, practice/drills-catalog.js, stats/predicted-prereqs-temp.js}* (frontend).